## TACFI Collaboration Agreement (AF) — Exos Aerospace

COLLABORATION AGREEMENT

BETWEEN

**AFRL/RQRI, Edwards AFB**

AND

Exos Aerospace Systems & Technologies, Inc.

FOR

TACTICAL FUNDS INCREASE (TACFI) PROGRAM EXECUTION

Version 1 - April 2026

**Authorities**

15 U.S.C. §638

U.S. Small Business Administration, Combined SBIR/STTR Policy Directive (https://www.sbir.gov/sites/default/files/SBA_SBIR_STTR_POLICY_DIRECTIVE_MAY_2023.pdf)

5 C.F.R. §2635.702(c), Exception (1)

Purpose: This collaboration agreement is intended to document interest and responsibilities from AFRL/RQRI, Edwards AFB and Exos Aerospace Systems & Technologies, Inc. in working with AFWERX and SpaceWERX Ventures and Exos Aerospace Systems & Technologies, Inc. on a project to be considered for additional Department of the Air Force (DAF) SBIR/STTR funding.

The Phase II agreement off of which this contemplated effort is based:

**FA9300-25-C-6022**

**SF233-0019**

**2 September 2025 – 2 October 2026**

**$1,897,824.01**

Was the Phase II from a solicitation by Open Topic or Specific Topic? Specific Topic (SF233-0019 — conventional Space Force topic; not Open Topic)

Is the Company considered a SBC and currently eligible for a TACFI SBIR/STTR award? Yes

Has the subject P2 effort been awarded a second (AKA “sequential”) P2 from any DoD service? No

Has the subject P2 effort ever been selected for or awarded a STRATFI or TACFI under a prior Notice of Opportunity? No

Is SBC executing a prior STRATFI or TACFI effort at the time of submission? No

Will anticipated work be performed in the United States? Yes

AFRL/RQRI and AFRL/RQRI has a national defense-related mission need in the area of responsive, reusable launch and advanced rocket (RDRE) propulsion — Speed and Reach of Disruption and Lethality.

5. Speed and Reach of Disruption and Lethality

>> SELECTED Strategic Capability: #5 Speed and Reach of Disruption and Lethality — RDRE detonation propulsion advances responsive, high-performance launch and boost propulsion.

This effort demonstrates support of the Operational Imperatives as indicated below:

4. Tactical Air Dominance: Supporting

6. Global Strike: Supporting

Core Competency: (Use only if this effort is Space-related, if unrelated, select N/A) The Space Technology should align with one of the six core competencies listed in the Space Capstone (The inaugural Space Capstone Publication, Spacepower (SCP)) (pg 33). This effort demonstrates support of the Core Competency as indicated below:

N/A — Air Force effort (Core Competency applies to Space Force efforts only).

The contemplated solution does support a Mission Area operational objective. This efforts capabilities have attributes that support the Mission Area as indicated below:

Mission Area 3 (MA3) Create the flexibility and mass to span a range of potential future crises and operate from the broader area that covers most of the world with positions resilient to limited adversary attack. Exos's reusable, recoverable platform enables RAPID test and iterative ADVANCEMENT of emerging propulsion and vehicle technologies — compressing the develop-test-fly cycle and accelerating future development across the physical world and the re-industrialization of space logistics — DIRECT: low-cost, reusable, rapidly dispersible launch (and proven in-space RDRE relight) creates the flexibility and MASS to operate from a globally distributed, resilient posture — Exos's cargo-class reusable space-logistics core thesis.

Impact of Implementing Solution: The Department needs responsive, low-cost, reusable launch and point-to-point delivery of mass effective from dispersed, contested, and austere locations, plus a rapid, recoverable means to mature next-generation rocket propulsion. Exos's new BLK3 — one of only three FAA-licensed reusable suborbital VTVL vehicles in the U.S. — integrated with a flight-weight Rotating Detonation Rocket Engine (RDRE; NASA printed design or GTL machined architecture) solves this by providing compact, high-performance, reusable propulsion and a 2-day-turnaround flight-test platform that characterizes new propulsion across the full altitude envelope (spaceport level to near-vacuum). We believe technology development under the subject SBIR/STTR topic may eventually contribute to solving a mission need. The main goals of our involvement in this project are to integrate the RDRE onto BLK3 and complete one hover and two recoverable flights to 85km+ over the 24-month period of performance, capturing altitude-swept RDRE performance data (detonation-wave stability vs. ambient backpressure, combustor wall heat flux, thrust oscillation amplitude/frequency vs. altitude, composite-airframe vibro-acoustic response) plus an in-space RDRE relight demonstration, with engine recovery and reuse between flights. The mission impact of this project on the DAF and the Department of Defense will be to provide the DAF the following strategic capability and otherwise strengthen the execution of the National Defense Strategy: a flight-qualified, low-cost, reusable detonation-propulsion capability — extensible to Orbital Transfer Vehicle (OTV) applications via proven in-space relight and to a reusable Black Brant-class point-to-point platform — that compresses the develop-test-fly cycle, accelerates fielding of advanced propulsion, and strengthens the Department's ability to generate and project combat power and mass from dispersed, resilient, globally distributed positions.

Project Timeline: Months 1-6 — RDRE integration onto the new BLK3; ground qualification (PDR-1 → CDR-1 → hot-fire checkouts); taxi/hover checkout (Hover 1). Months 7-18 — Flight 1 to 85km+ (Spaceport America), recovery, RDRE refurbishment; Flight 2 to 85km+ including in-space relight demonstration. Months 18-24 — data reduction, OTV/relight analysis, final report. (1 hover + 2 flights to 85km+ over the 24-month period of performance.)

(Note: Failure to meet prescribed timelines will result in a selected effort being placed at risk for funding consideration.)

If the SBIR/STTR “TACFI” effort is selected, this project will be awarded within 90 days after notification of selection for funding for a duration of 24 months.

Failure to award subject effort within prescribed timelines may result in forfeiture of supplemental funding. By signing this agreement, it is acknowledged that an award timeline of more than 90 days is not acceptable and will result in potential loss of SBIR funds.

Stakeholders: We understand the successful transition of innovative solutions requires multiple engaged and empowered stakeholders. We have coordinated with, assigned responsibilities to, and received concurrence from, the following stakeholders relative to this effort:

Primary DAF End-User Organization: AFRL/RQRI (Rocket Propulsion, Edwards AFB) is the single primary End-User. As the DoD center of excellence for rocket propulsion and the managing organization for the Phase II RDRE effort (FA9300-25-C-6022), AFRL/RQRI is the direct technical owner and transition customer for RDRE maturation and benefits operationally from the solution.

**Technical Directorate (TD):**

**Program Executive Office (PEO):**

**Major Command (MAJCOM):**

Primary End-User Points of Contact (POC): (required/do not appoint a GO) Dr. Matthew Harvazinski, Civilian (Dr.), Program Manager, matthew.harvazinski.1@spaceforce.mil, AFRL/RQRI

**Alternate End-User POC: (required/do not appoint a GO) [GOV POC — AFRL/RQRI to designate]**

**Additional Organizations: N/A**

Primary Customer Organization: AFRL/RQRI is the single primary Customer / procuring activity, responsible for acquiring the solution on behalf of the End-User; it is capable of integrating and transitioning the technology and is supported by contracting at AFTC/PZRB, Edwards AFB.

Which acquisition community will procure and/or support this solution, and why are they the right selection?

If you do not identify a Program Acquisition Executive (PAE)/Program Executive Office(PEO), why is this organization appropriate?

**Technical Directorate (TD):**

**Program Executive Office (PEO):**

**Major Command (MAJCOM):**

**Primary Customer POC: (required/do not appoint a GO) [GOV POC — AFRL/RQRI to designate]**

**Alternate Customer POC: (required/do not appoint a GO) [GOV POC — AFRL/RQRI to designate]**

**Additional Organizations: N/A**

**Primary TPOC: (required/do not appoint a GO) Jason Burr, Technical POC, jason.burr.5@us.af.mil, AFRL/RQRI**

Alternate TPOC: (required/do not appoint a GO) Dr. Matthew Harvazinski, matthew.harvazinski.1@spaceforce.mil, AFRL/RQRI

TPOCs will work closely with the company during TACFI execution to achieve project success. This includes, but is not limited to: receiving and accepting/rejecting deliverables; providing feedback to the company on the solution; overcoming tactical barriers to TACFI execution (e.g. ATO, Base Access, Waivers/Certifications, etc.); and driving the effort towards Phase III transition.

**SIPR POC/Email: (required) [GOV POC — AFRL/RQRI to provide SIPR POC]**

**Finance (required): [GOV POC — AFRL/RQRI finance office]**

Contracting Officer (required): AFTC/PZRB, Kiera Joyner, Contracting Officer, kiera.joyner@us.af.mil. This POC will be sent automated status emails - please ensure the correct and appropriate name, grade, email, etc is identified.

**Contracting Officer's Supervisor (required): [GOV POC — AFTC/PZRB]**

**Legal (required): [GOV POC — AFTC/AFRL legal office]**

**Information Assurance/Cybersecurity: :**

**Security Office:**

**Managing Organization Program Manager: [GOV POC — AFRL/RQRI SBIR/STTR PM, if applicable]**

If this effort is selected for funding, AFTC/PZRB (Edwards AFB) will assume responsibility for drafting and releasing a request for proposal (RFP) and awarding the TACFI contract or Fund Increase modification on the existing Phase 2 (or delegating as appropriate).

N/A — Not applicable. The DAF Contracting Office (AFTC/PZRB) identified above will execute this effort; no non-DAF contracting office or exemption is requested.

**Transition Strategy (vendor response):**

• Phase III Funding: Following TACFI completion, Exos transitions via sole-source Phase III under 15 U.S.C. §638(r), citing FA9300-25-C-6022 — no further competition required. Funding sources: USAF/USSF program-of-record dollars, NASA Flight Opportunities task orders, and Exos Series A capital.

• Contract Vehicle / Contracting Org: Sole-source Phase III contract or sequential PIIb modification executed by AFTC/PZRB (Edwards AFB); Contracting Officer Kiera Joyner.

• Requirements Alignment: Aligns with USAF Operational Imperatives (Global Strike, Tactical Air Dominance) and USSF Space Mobility & Logistics; End-User AFRL/RQRI holds a validated rocket-propulsion maturation need.

• Integration / Scalability: Flight-qualified RDRE integrates onto boost-propulsion and responsive/reusable launch and scales to Orbital Transfer Vehicles (OTVs) and a reusable Black Brant-class point-to-point platform; technology owned by Exos (with GTL/NASA licensing as applicable); scalable across services and ranges.

• Commercialization: Exos is one of only three FAA-licensed reusable suborbital providers and sells flight-test/payload services today; RDRE de-risking opens commercial OTV and responsive-launch markets.

• Post-SBIR/STTR Testing: Continued recoverable flight testing (2-day turnaround) for RDRE qualification and in-space relight demonstration.

Installation Access: The SBIR/STTR Company will not require access to military installations to perform the contemplated work under this effort.

If they will require CAC/Base Access: N/A — no military installation access required; work is performed at Exos's Greenville, TX facility and commercial launch ranges (e.g., Spaceport America).

Cyber Security: Nearly all software and information technology (IT) solutions that integrate with DAF information systems, weapons systems, or DAF data will require an Authority to Operate (ATO) from the owning Authorization Official (AO). The AO is a senior DAF official who is responsible for cybersecurity risks to the information/weapon system. This approval involves a thorough Risk Management Framework (RMF) assessment, and close collaboration between the AO, Government customer, and company. The RMF incorporates the cybersecurity control requirements outlined in NIST SP 800-53, 800-171, & 800-172 (where applicable). The DAF TPOC or Customer MUST consult with their applicable wing-level cybersecurity or information assurance office to determine whether an ATO is required. If an ATO is required, you may coordinate with DAF Software Factories, such as Platform One, which offer streamlined ATO processes. Most of these pathways will require additional funding from the DAF customer to the AO or Software Factory to cover ATO-related costs.

The contemplated solution does not involve software or Information Technology.

N/A — an ATO is not required (the solution does not involve software or Information Technology); see below.

AFRL/RQRI has confirmed with its Information Assurance/Cybersecurity office that an ATO will not be required to field this solution.

**Flight Testing or Utilization of Unmanned Aerial Systems (UAS):**

The SBIR/STTR Company will require flight testing (suborbital flights of the BLK3 reusable vehicle) to accomplish the contemplated work under this effort. The effort does not utilize UAS.

The contemplated work under this SBIR/STTR TACFI effort is not focused on Counter-Unmanned Aircraft Systems (cUAS).

Human Subject Research: Human Research Protection Office (HRPO) review is required for all activities which are or might include Human Subjects Research that is supported by a DoD institution. Support may include anything from providing funding to an institution that is actually carrying out the research (such as through a contract or similar arrangement), to providing direct resources, access to study subjects or their data, etc. Most often these activities are reviewed and approved by an outside Institutional Review Board or Exempt Determination Office.

Human Subject Research (HSR) will not be conducted to accomplish the contemplated work under this effort.

HRPO: N/A (no Human Subject Research)

**Approval Signatures**

**DAF End-User Champion Customer Champion**

_____ _____

Signature Signature

**Name, Rank/GS Equivalent, Title Name, Rank/GS Equivalent, Title** _____ _____

(Date) (Date)